3.1 - Integration: IBM i, Boomi, Astea

Integration Between IBM i, Astea, and Boomi
Integrating IBM i, Astea, and Boomi (Dell Boomi) involves connecting on-premises legacy systems (IBM i), a field service management solution (Astea), and a cloud-based integration platform (Boomi).
This type of integration is used to streamline field service operations, synchronize customer and work order data, and automate workflows between IBM i and Astea.

1. Understanding Each Component

IBM i

- Hosts ERP systems, inventory management, financial applications, and custom RPG programs.
- Uses IBM Db2 for i as its database.
- Communicates via SOAP APIs.

Astea

- A cloud-based or on-prem Field Service Management (FSM) platform.
- Manages work orders, dispatching, technician schedules, inventory tracking, and customer service.
- Provides APIs for integration with ERPs, CRMs, and third-party tools.

Boomi (iPaaS - Integration Platform as a Service)

- Connects IBM i and Astea via pre-built connectors.
- Uses event-driven integration to trigger workflows automatically.
- Supports API Management, ETL, B2B/EDI, and workflow automation.

2. Integration Architecture

The integration can be set up in multiple ways, depending on the use case.
A. Data Flow - IBM i → Boomi → Astea

1. IBM i generates or updates data (e.g., customer info, work orders, inventory).
2. Boomi extracts data via:

    - JDBC Connector (IBM Db2 for i)
    - REST/SOAP API Calls (if IBM i exposes web services)
    - MQ, FTP, or CSV File Transfers
3. Boomi transforms the data into a format Astea can consume.
4. Boomi pushes data to Astea using:
    - Astea API
    - Database inserts (if Astea is on-prem and allows direct access)

B. Data Flow - Astea → Boomi → IBM i

1. Astea captures new field service data (e.g., completed work orders, technician updates).
2. Boomi pulls the data via:
    - Astea's REST/SOAP APIs
    - Webhooks (event-driven)
    - Database queries (if on-prem)
3. Boomi transforms and maps the data to IBM i format.
4. Boomi inserts data into IBM i using:
    - JDBC (IBM Db2 for i)
    - Web Services (REST/SOAP)
    - File-based integration (FTP, MQ, CSV drop)

3. Key Use Cases for Integration

- Customer Data Sync
- Ensure customer records in IBM i match Astea's customer database.
- Real-time or scheduled sync via Boomi API Calls & Db2 queries.
- Work Order Management
- When a new work order is created in IBM i, sync it to Astea.
- When a technician closes a work order in Astea, update IBM i's ERP.
- Inventory & Parts Management
- Synchronize inventory levels between IBM i and Astea.
- If a technician uses a part, update IBM i's inventory system.
- Billing & Invoicing
- After service completion in Astea, push billing data to IBM i's ERP.
- Automate invoice generation and financial reconciliation.

- Technician & Scheduling Updates
- Fetch technician schedules from Astea and sync them with IBM i.
- Send real-time job updates to IBM i when a technician accepts a task.

4. Technical Considerations

- API vs. Direct Database Connection
- If Astea is cloud-based, use Boomi's REST API Connector.
- If Astea is on-prem, use direct JDBC database queries for faster access.
- Data Transformation & Mapping
- IBM i often uses fixed-length fields or EBCDIC encoding, while Astea uses modern UTF-8 JSON.
- Boomi handles ETL (Extract, Transform, Load) and converts formats.
- Real-Time vs. Batch Processing
- Boomi Webhooks enable real-time updates from Astea.
- Batch jobs (Boomi schedules) process large volumes overnight.
- Error Handling & Logging
- Boomi provides error tracking and logging for API failures.
- Implement retries & alerting mechanisms for failed transactions.

5. Example Boomi Workflow

Scenario: Work Order Update from Astea to IBM i using SOAP
Since Astea is using SOAP Web Services, Boomi will communicate with Astea's SOAP API instead of a REST API. The workflow will handle XML payloads, including SOAP envelopes and WSDL-based communication.

Step-by-Step Workflow (SOAP-Based)

1. Astea generates a work order update
    - Astea's system exposes a SOAP Web Service to send work order updates.
    - The SOAP request is structured in XML format, following the WSDL definition.
2. Boomi pulls the work order update from Astea via SOAP
   Boomi uses its SOAP Client Connector to send a request to Astea's

SOAP Web Service.
The request might include a timestamp filter to fetch only recent updates.

Example SOAP request to Astea:xml

```xml
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ast="http://www.astea.com/">
  <soapenv:Header/>
  <soapenv:Body>
    <ast:GetWorkOrderUpdates>
      <ast:LastUpdated>2025-03-04T12:00:00Z</ast:LastUpdated>
    </ast:GetWorkOrderUpdates>
  </soapenv:Body>
</soapenv:Envelope>
```

  - Astea responds with an XML payload containing the updated work order.

1. Boomi processes and transforms the SOAP response
   - The response is parsed using Boomi’s XML profile.
   - Data is transformed into IBM i's expected format (e.g., fixed-length records, ASCII to EBCDIC conversion if needed).

Example response from Astea:xml

```xml
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Body>
    <GetWorkOrderUpdatesResponse xmlns="http://www.astea.com/">
      <WorkOrder>
        <ID>12345</ID>
        <Status>Completed</Status>
        <Technician>John Doe</Technician>
        <CompletionDate>2025-03-04T15:30:00Z</CompletionDate>
      </WorkOrder>
    </GetWorkOrderUpdatesResponse>
```

```xml
  </soapenv:Body>
</soapenv:Envelope>
```

1. Boomi updates IBM i system via SOAP or Database Connection
   - If IBM i exposes a SOAP Web Service, Boomi sends a SOAP request to update the work order in IBM i.
   - If IBM i does not use SOAP, Boomi writes the data directly to IBM Db2 for i (AS/400 database) via JDBC.

   Example SOAP request to IBM i:xml

```xml
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/
soap/envelope/"
                  xmlns:ibm="http://www.ibm.com/">
  <soapenv:Header/>
  <soapenv:Body>
    <ibm:UpdateWorkOrder>
      <ibm:ID>12345</ibm:ID>
      <ibm:Status>Completed</ibm:Status>
      <ibm:Technician>John Doe</ibm:Technician>
      <ibm:CompletionDate>2025-03-04T15:30:00Z</
ibm:CompletionDate>
    </ibm:UpdateWorkOrder>
  </soapenv:Body>
</soapenv:Envelope>
```

   - If using JDBC, Boomi executes an SQL UPDATE query instead.
1. Boomi handles success/failure logging
   - Boomi receives a response from IBM i’s SOAP service or database.
   - If the update is successful, Boomi logs the confirmation.
   - If the update fails, Boomi:
     - Retries the request (configurable retry logic).

    - Sends an alert/email if multiple attempts fail.
    - Logs the error in a centralized monitoring system.

Alternative: IBM i to Astea (SOAP-Based Push)

- If IBM i needs to push data to Astea, it can expose a SOAP Web Service.
- Boomi can send SOAP requests from IBM i to Astea in the same way.
- Example use case:
    - A new customer is added in IBM i, and Boomi calls Astea's SOAP API to sync the customer record.

Key Adjustments for SOAP-Based Integration

- Use SOAP Client Connector in Boomi instead of REST.
- Handle XML transformation and mapping (instead of JSON).
- Ensure correct SOAP headers & namespaces (matching WSDL specs).
- Use WSDL import for automatic SOAP message structure validation.
- Enable fault handling (SOAP Faults vs. HTTP errors).

Integrating IBM i, Astea, and Boomi modernizes field service operations by ensuring smooth data synchronization, automation, and real-time updates. Boomi acts as the middleware, allowing seamless communication between legacy ERP systems (IBM i) and modern cloud-based applications (Astea).

6. Benefits of Boomi for IBM i & Astea Integration

- Automates manual data entry, reducing errors.
- Improves data accuracy between IBM i and Astea.
- Enables real-time field service updates for technicians.
- Scales easily as business needs grow.
- Simplifies maintenance with Boomi's low-code integration tools.